Statistics for Data Science - II
Graded Questions
Jan 2025 Term
Week-6

Question

1. Let the random variables X and Y be uniformly distributed in the triangular region D, where D is bounded by $y = 0$, $x = 3y$, and $x = 3$.

(i). Find the joint PDF of X and Y.

(a) $f_{XY}(x, y) =$
$$\begin{cases} \frac{2}{3}, & 0 \le x \le 3, \quad 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(b) $f_{XY}(x, y) =$
$$\begin{cases} \frac{3}{2}, & 3y \le x \le 3, \quad 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) $f_{XY}(x, y) =$
$$\begin{cases} \frac{3}{2}, & 0 \le x \le 3y, \quad 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(d) $f_{XY}(x, y) =$
$$\begin{cases} \frac{2}{3}, & 3y \le x \le 3, \quad 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Answer: (d)

Solution:

Let the random variables X and Y be uniformly distributed in the triangular region D, where D is bounded by:

- $y = 0$ (the x-axis)
- $x = 3y$ (a line through the origin with slope 3)
- $x = 3$ (a vertical line at $x = 3$)

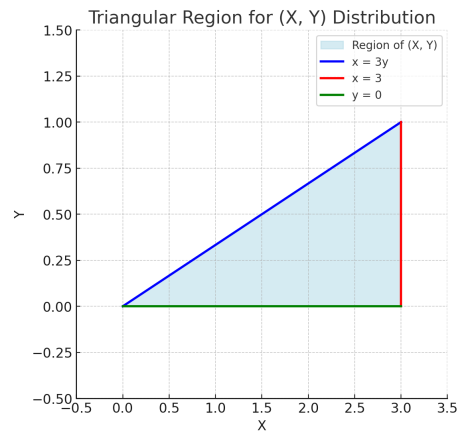

Figure 1: Triangular region bounded by $x = 3y$, $x = 3$, and $y = 0$.

Step 1: Determining the Area of Region D

The region D forms a right triangle with:

- Base $= 3$
- Height $= 1$

Thus, the area of D is:

$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height} = \frac{1}{2} \times 3 \times 1 = \frac{3}{2}.$$

Step 2: Formula for Joint PDF

Since X and Y are uniformly distributed over D, the joint PDF is given by:

$$f_{XY}(x, y) = \frac{1}{\text{Area of } D}, \quad (x, y) \in D.$$

Substituting the area:

$$f_{XY}(x, y) = \frac{1}{\frac{3}{2}} = \frac{2}{3}, \quad (x, y) \in D.$$

Step 3: Finding the Correct Limits

Observing the given region:

- Y ranges from 0 to 1. - For a fixed y, X varies from $3y$ to 3, i.e., $3y \leq x \leq 3$.

Thus, the correct joint PDF is:

$$f_{XY}(x,y) = \begin{cases} \frac{2}{3}, & 3y \leq x \leq 3, \quad 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Step 4: Selecting the Correct Answer

From the given options:

(a) $f_{XY}(x,y)$ has incorrect limits for x.

(b) The coefficient $\frac{3}{2}$ is incorrect.

(c) The limits for x do not match the given region.

(d) Correctly represents the given region and has the correct density value.

Final Answer:

$$\text{(d)} \quad f_{XY}(x,y) = \begin{cases} \frac{2}{3}, & 3y \leq x \leq 3, \quad 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(ii) Find $f_{X|Y=y}(x)$

$$(a) \quad f_{X|Y=y}(x) = \begin{cases} \frac{1}{3(1-y)}, & 3y \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

$$(b) \quad f_{X|Y=y}(x) = \begin{cases} \frac{2}{3(1-y)}, & 3y \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

$$(c) \quad f_{X|Y=y}(x) = \begin{cases} 2(1-y), & 3y \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

$$(d) \quad f_{X|Y=y}(x) = \begin{cases} 3(1-y), & 3y \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Answer: (a)

Solution:

Finding $f_{X|Y=y}(x)$

Given that the joint probability density function (PDF) of X and Y is:

$$f_{XY}(x,y) = \begin{cases} \frac{2}{3}, & 3y \le x \le 3, \quad 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

The conditional PDF of X given $Y = y$, denoted as $f_{X|Y=y}(x)$, is found using the formula:

$$f_{X|Y=y}(x) = \frac{f_{XY}(x,y)}{f_Y(y)}$$

where $f_Y(y)$ is the marginal PDF of Y, computed as:

$$f_Y(y) = \int_{3y}^{3} f_{XY}(x,y)\,dx.$$

Substituting $f_{XY}(x,y) = \frac{2}{3}$:

$$f_Y(y) = \int_{3y}^{3} \frac{2}{3}\,dx = \frac{2}{3}(3-3y) = \frac{2}{3} \cdot 3(1-y) = 2(1-y).$$

Now, computing $f_{X|Y=y}(x)$:

$$f_{X|Y=y}(x) = \frac{\frac{2}{3}}{2(1-y)} = \frac{1}{3(1-y)}, \quad \text{for } 3y \le x \le 3.$$

Thus, the correct answer is:

$$f_{X|Y=y}(x) = \begin{cases} \frac{1}{3(1-y)}, & 3y \le x \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

Final Answer:

$$\text{(a)} \quad f_{X|Y=y}(x) = \begin{cases} \frac{1}{3(1-y)}, & 3y \le x \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

Question

2. At a particular bakery, dough is prepared in bulk each day. Let a random variable X denote the amount of dough prepared in a given day, and let Y denote the amount of dough used to bake items in the same day (in some normalized units). The bakery cannot use more dough than what was prepared in a given day.

Assume the joint density function of X and Y is given by:

$$f_{XY}(x,y) = \begin{cases} cxy, & 0 \le y \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability that the amount of dough used to bake items in a given day is less than half the amount prepared on that day. Enter the answer correct to two decimal places.

Answer: 0.25, **Range:** 0.22 to 0.28

Solution:

Step 1: Determine the Constant c

The joint probability density function $f_{XY}(x,y)$ must integrate to 1 over its entire domain:

$$\int_0^2 \int_0^x cxy \, dy \, dx = 1.$$

Evaluating the inner integral:

$$\int_0^x cxy \, dy = cx \int_0^x y \, dy = cx \left[\frac{y^2}{2} \right]_0^x = cx \cdot \frac{x^2}{2} = \frac{cx^3}{2}.$$

Now, integrating with respect to x:

$$\int_0^2 \frac{cx^3}{2} \, dx = \frac{c}{2} \int_0^2 x^3 \, dx.$$

$$= \frac{c}{2} \left[\frac{x^4}{4} \right]_0^2 = \frac{c}{2} \cdot \frac{2^4}{4} = \frac{c}{2} \cdot \frac{16}{4} = \frac{4c}{2} = 2c.$$

Setting this equal to 1:

$$2c = 1 \quad \Rightarrow \quad c = \frac{1}{2}.$$

Step 2: Compute the Required Probability

We need to find:

$$P(Y < X/2) = \int_0^2 \int_0^{x/2} f_{XY}(x,y) \, dy \, dx.$$

Substituting $f_{XY}(x,y) = \frac{1}{2}xy$:

$$P(Y < X/2) = \int_0^2 \int_0^{x/2} \frac{1}{2} xy \, dy \, dx.$$

Evaluating the inner integral:

$$\int_0^{x/2} \frac{1}{2} xy \, dy = \frac{x}{2} \int_0^{x/2} y \, dy.$$

$$= \frac{x}{2} \left[\frac{y^2}{2} \right]_0^{x/2} = \frac{x}{2} \cdot \frac{(x/2)^2}{2} = \frac{x}{2} \cdot \frac{x^2}{4} \cdot \frac{1}{2} = \frac{x^3}{16}.$$

Now, integrating with respect to x:

$$\int_0^2 \frac{x^3}{16} \, dx = \frac{1}{16} \int_0^2 x^3 \, dx.$$

$$= \frac{1}{16} \left[\frac{x^4}{4} \right]_0^2 = \frac{1}{16} \cdot \frac{2^4}{4} = \frac{1}{16} \cdot \frac{16}{4} = \frac{4}{16} = \frac{1}{4}.$$

Final Answer

$$P(Y < X/2) = 0.25.$$

Thus, the probability that the amount of dough used is less than half the amount prepared on that day is **0.25**.